ANNEXURE A

DIGITAL IMAGE PROCESSING –

SCENARIO-BASED ANSWERS

Q1. Image Enhancement Using Grey-Level Transformations

1. Problem Statement

A hospital uses a digital imaging system to capture X-ray images, but some images appear too dark to identify important anatomical structures. Grey-level transformation techniques are applied to improve the brightness and contrast of the images.

2. Objective

1. To improve the brightness of dark X-ray images.
2. To enhance contrast between anatomical structures.
3. To calculate transformed pixel intensity values.
4. To improve the visibility of important features in medical images.

3. Theory

Grey-level transformation is an image enhancement technique that converts an input pixel intensity into a new output intensity.

Formula: $s = T(r)$

Where:

- r = Input pixel intensity
- s = Output pixel intensity
- $T(r)$ = Transformation function

For an 8-bit grayscale image, the intensity range is 0–255.

- 0 represents black.
- 255 represents white.

- Intermediate values represent different shades of grey.

4. Types of Grey-Level Transformations

1. Negative Transformation

Formula: $s = 255 - r$

This transformation reverses the intensity values of an image. Dark pixels become bright, and bright pixels become dark.

Example:

If $r = 50$,

$s = 255 - 50$

$s = 205$

Therefore, the output intensity is 205.

2. Logarithmic Transformation

Formula: $s = c \times \log(1 + r)$

Where c is a scaling constant.

This transformation expands darker intensity values and compresses brighter intensity values. It is useful when details in dark regions are difficult to observe.

3. Gamma Transformation

Formula: $s = 255 \times (r / 255)^{\gamma}$

Where:

- r = Input intensity
- s = Output intensity
- γ = Gamma correction parameter

If γ is less than 1, the image becomes brighter.

If γ is greater than 1, the image becomes darker.

Gamma transformation is suitable for enhancing underexposed X-ray images.

4. Contrast Stretching

Formula:

$s = [(r - r_{min}) / (r_{max} - r_{min})] \times 255$

Where:

- r_{min} = Minimum input intensity
- r_{max} = Maximum input intensity

Contrast stretching expands the range of intensity values and improves image contrast.

5. Step-by-Step Calculation

Consider a dark X-ray image. Apply gamma transformation with $\gamma = 0.5$.

Formula:

$s = 255 \times (r / 255)^{0.5}$

Example 1: Input intensity = 16

Step 1: Substitute the input value.

$s = 255 \times (16 / 255)^{0.5}$

Step 2: Simplify.

$s = 255 \times \sqrt{(16 / 255)}$

Step 3: Calculate.

$s \approx 64$

Final answer: Output intensity = 64.

Example 2: Input intensity = 64

Step 1: Substitute the input value.

$s = 255 \times (64 / 255)^{0.5}$

Step 2: Simplify.

$s = 255 \times \sqrt{(64 / 255)}$

Step 3: Calculate.

$s \approx 128$

Final answer: Output intensity = 128.

Example 3: Input intensity = 144

Step 1: Substitute the input value.

$s = 255 \times (144 / 255)^{0.5}$

Step 2: Simplify.

$s = 255 \times \sqrt{(144 / 255)}$

Step 3: Calculate.

$s \approx 192$

Final answer: Output intensity = 192.

6. Results

Input intensity	Output intensity
16	64
64	128
144	192
225	240

The results demonstrate that gamma transformation with $\gamma = 0.5$ increases the brightness of darker pixels more significantly than brighter pixels.

7. Explanation of Image Enhancement

Grey-level transformations improve X-ray image quality in the following ways:

1. Brightness enhancement makes dark anatomical structures easier to observe.
2. Contrast enhancement helps distinguish neighbouring tissues with different intensities.

3. Improved visibility supports the examination of important anatomical features.
4. Appropriate intensity mapping improves the visual interpretation of the image.

However, excessive enhancement can amplify noise. Therefore, the original image should be preserved, and the enhanced image should be validated before clinical use.

8. Image for Q1

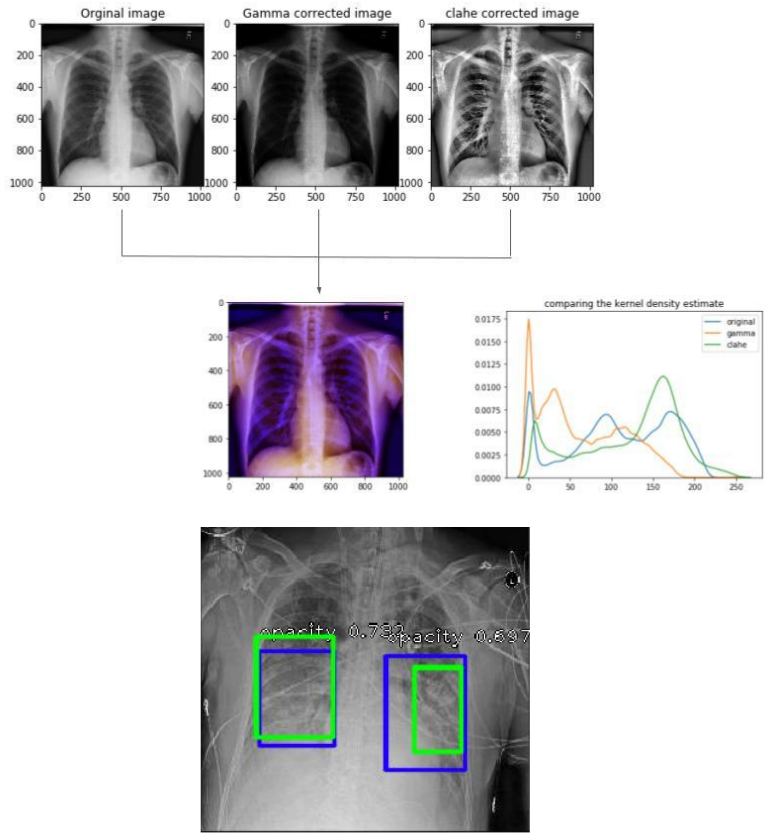


Figure 1: Illustration of original dark image, gamma transformation and contrast stretching.

9. Conclusion

Grey-level transformations improve the brightness and contrast of dark X-ray images. Gamma transformation with γ less than 1 is suitable for brightening underexposed images, while contrast stretching improves the use of the available intensity range. These techniques help make important anatomical features more visible.

Q2. Histogram Processing and Spatial Filtering

1. Problem Statement

A security surveillance system captures images at night, resulting in poor contrast and unwanted noise. The system must enhance the images to identify people and objects more clearly. Histogram equalization and spatial filtering are applied to improve image quality.

2. Objective

1. To improve image contrast using histogram equalization.
2. To reduce noise using spatial filtering.
3. To preserve important image details.
4. To compare the effects of both techniques.

3. Theory

Histogram processing studies the distribution of pixel intensity values in an image.

A histogram represents the number of pixels corresponding to each intensity level.

For an 8-bit grayscale image, the intensity values range from 0 to 255.

If most pixels occupy a narrow intensity range, the image may appear dark or have poor contrast.

Histogram equalization redistributes the intensity values to improve the overall contrast of the image.

4. Histogram Equalization

The probability of an intensity level is calculated using the following formula.

$p(r_k) = n_k / N$

Where:

- n_k = Number of pixels with intensity r_k
- N = Total number of pixels

The cumulative distribution function is:

5. Step-by-Step Numerical Example

Consider a small image containing eight pixels with four intensity levels: 0, 1, 2 and 3.

Intensity	Pixel count	Probability	CDF
0	4	0.50	0.50
1	2	0.25	0.75
2	1	0.125	0.875
3	1	0.125	1.00

Since L = 4:

$s_k = \text{Round}[3 \times \text{CDF}(r_k)]$

Example 1: Intensity = 0

$s = \text{Round}[3 \times 0.50]$

$s = \text{Round}[1.50]$

$s = 2$

Example 2: Intensity = 1

$s = \text{Round}[3 \times 0.75]$

$s = \text{Round}[2.25]$

$s = 2$

Example 3: Intensity = 2

$s = \text{Round}[3 \times 0.875]$

$s = \text{Round}[2.625]$

$s = 3$

Example 4: Intensity = 3

$s = \text{Round}[3 \times 1.00]$

$s = 3$

The transformed intensity values are 2, 2, 3 and 3.

This simplified example demonstrates the mapping calculation. Actual contrast improvement depends on the image's original histogram.

6. Spatial Filtering

Spatial filtering processes pixels using their neighbouring pixels.

A. Mean Filter

The mean filter replaces a pixel with the average of the values in its neighbourhood.

For a 3 × 3 neighbourhood:

Output pixel = Sum of the 9 neighbouring pixel values / 9

It reduces some random noise but can blur edges.

B. Median Filter

The median filter replaces a pixel with the median of its neighbouring values.

Consider the following pixel values:

11, 12, 12, 13, 14, 14, 15, 16, 250

The median value is:

Median = 14

The extreme noise value 250 is replaced by 14.

Therefore, the median filter is particularly effective for removing salt-and-pepper noise.

7. Comparison of Techniques

Feature	**Histogram equalization**	**Spatial filtering**
Main purpose	Improve contrast	Reduce noise
Processing method	Intensity remapping	Neighbourhood operations
Effect on brightness	Changes intensity distribution	Depends on filter
Edge preservation	Does not directly preserve edges	Depends on filter type

Example	Histogram equalization	Median filter

8. Application to Surveillance Images

The recommended processing procedure is:

1. Convert the captured image to grayscale if required.
2. Apply histogram equalization to improve contrast.
3. Apply a median filter to remove isolated noisy pixels.
4. Examine the resulting image for visible objects and boundaries.
5. Use CLAHE if global equalization amplifies noise excessively.

9. Image for Q2

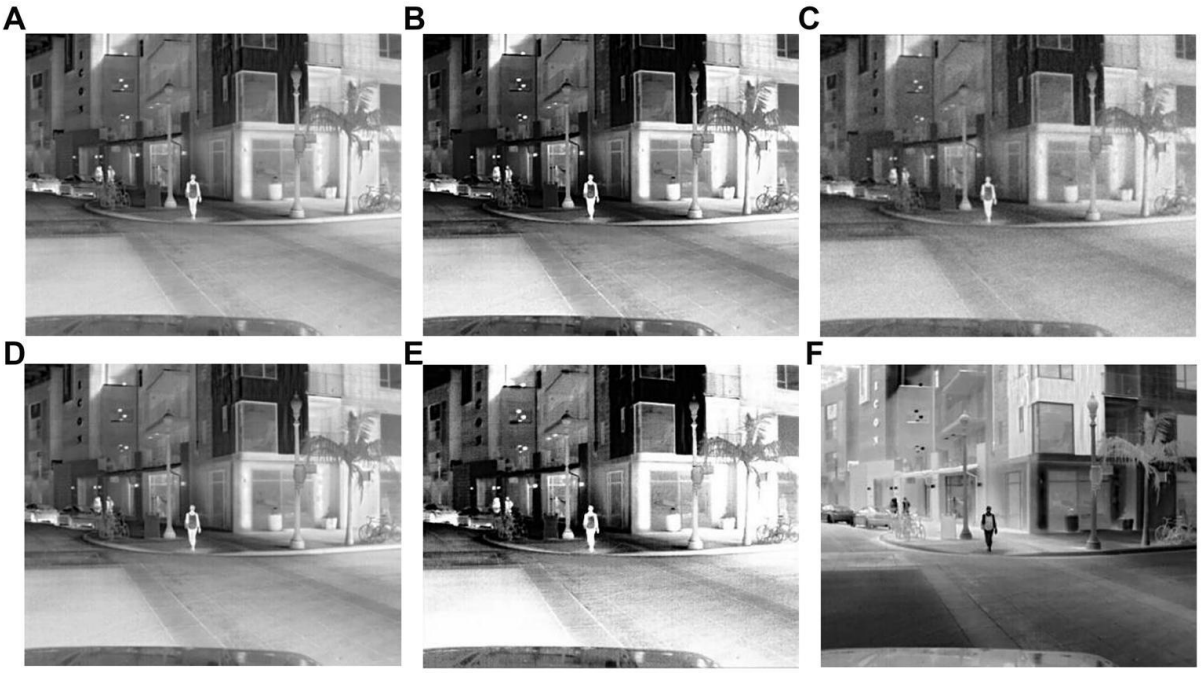


Figure 2: Illustration of original nighttime image, histogram equalization and spatial noise filtering.

10. Conclusion

Histogram equalization improves image contrast, while spatial filtering reduces unwanted noise. Combining these techniques helps surveillance systems identify people and objects more clearly. The filter must be selected carefully to avoid losing important details.

Q3. Image Smoothing and Sharpening Using Frequency Domain Filtering

1. Problem Statement

A satellite imaging system captures land images containing noise and blurred boundaries between different geographical regions. Frequency domain filtering is used to improve image quality by reducing noise and sharpening important geographical features.

2. Objective

1. To understand frequency domain filtering.
2. To reduce high-frequency noise using a low-pass filter.
3. To sharpen edges using a high-pass filter.
4. To compare the effects of smoothing and sharpening.

3. Theory

An image contains different spatial frequencies.

- Low frequencies represent slowly changing intensity patterns.
- High frequencies represent rapid intensity changes, including edges and noise.

The Fourier transform converts an image from the spatial domain into the frequency domain.

F(u,v) = Fourier Transform of f(x,y)

The filtering operation is:

$G(u,v) = H(u,v) \times F(u,v)$

Where:

- F(u,v) = Original frequency spectrum
- H(u,v) = Filter transfer function
- G(u,v) = Filtered frequency spectrum

The final image is reconstructed using the inverse Fourier transform.

g(x,y) = Inverse Fourier Transform of G(u,v)

4. Low-Pass Filtering

A low-pass filter allows low-frequency components to pass while attenuating high-frequency components.

A Gaussian low-pass filter is defined as:

$H_LP(u,v) = e^{[-D^2(u,v) / (2D_0^2)]}$

Where:

- $D(u,v)$ = Distance from the frequency origin
- D_0 = Cutoff frequency

Procedure:

1. Convert the image into the frequency domain.
2. Construct the low-pass filter.
3. Multiply the frequency spectrum by the filter.
4. Apply the inverse Fourier transform.
5. Examine the smoothed image.

Advantages:

- Reduces high-frequency noise.
- Smooths uniform land regions.
- Reduces small intensity fluctuations.

Limitation: It may blur narrow roads, rivers and geographical boundaries.

5. High-Pass Filtering

A high-pass filter attenuates low-frequency components while retaining high-frequency components.

Gaussian high-pass filter:

$H_HP(u,v) = 1 - H_LP(u,v)$

It emphasizes rapid intensity changes and highlights boundaries.

However, high-pass filtering can amplify noise and may produce an edge-dominated image.

6. High-Boost Sharpening

High-boost filtering combines the original image with its high-frequency details.

7. Step-by-Step Numerical Example

Suppose:

Original pixel intensity = 150

Low-pass filtered intensity = 120

Sharpening factor k = 1.5

Step 1: Apply the formula.

Output = 150 + 1.5 × (150 − 120)

Step 2: Calculate the difference.

Output = 150 + 1.5 × 30

Step 3: Multiply.

Output = 150 + 45

Final answer: Output intensity = 195.

This result assumes the sharpened value remains within the valid intensity range of the image.

8. Comparison of Filters

Feature	**Low-pass filter**	**High-pass filter**
Main purpose	Smoothing	Edge enhancement
Frequency response	Retains low frequencies	Retains high frequencies
Noise effect	Reduces some high-frequency noise	May amplify noise

Edge appearance	Softer edges	More prominent edges
Application	Noise reduction	Boundary enhancement

9. Application to Satellite Images

The recommended processing sequence is:

1. Apply low-pass filtering to reduce noise.
2. Inspect whether important boundaries have become blurred.
3. Apply moderate high-boost sharpening.
4. Compare the output with the original image.
5. Select suitable filter parameters to preserve meaningful geographical features.

10. Image for Q3

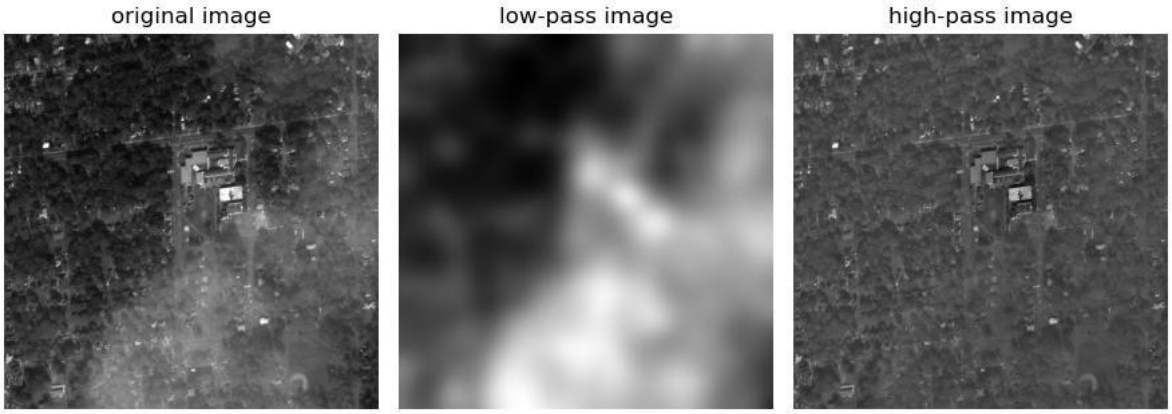


Figure 3: Illustration of original noisy satellite image, low-pass smoothing and high-boost sharpening.

11. Conclusion

Low-pass filtering reduces high-frequency noise and smooths an image. High-pass filtering enhances rapid intensity changes and highlights boundaries. Combining controlled smoothing and sharpening improves satellite image interpretation, although excessive filtering may remove useful details or amplify noise.

Q4. Image Segmentation Using Edge Detection and Thresholding

1. Problem Statement

An automated manufacturing system inspects metal components to identify cracks and surface defects. Variations in brightness make defect detection difficult. Edge detection and thresholding are applied to identify boundaries and separate defective regions from the background.

2. Objective

1. To identify possible crack boundaries.
2. To detect defective regions.
3. To generate a binary defect image.
4. To improve automated quality inspection.

3. Theory

Image segmentation divides an image into meaningful regions.

In manufacturing inspection, segmentation separates defective regions from normal material.

The two main techniques used are:

1. Edge detection.
2. Thresholding.

Edge detection identifies abrupt intensity changes, while thresholding classifies pixels according to their intensity values.

4. Edge Detection Using the Sobel Operator

The Sobel operator estimates horizontal and vertical intensity gradients.

Horizontal Sobel kernel:

Gx =

[−1 0 1]

[−2 0 2]

[−1 0 1]

Vertical Sobel kernel:

Gy =

[−1 −2 −1]

[0 0 0]

[1 2 1]

The gradient magnitude is calculated as:

$M(x,y) = \sqrt{(Gx^2 + Gy^2)}$

A large gradient magnitude indicates a possible edge.

5. Step-by-Step Numerical Example

Suppose the gradient values are:

Gx = 3

Gy = 4

Step 1: Apply the gradient formula.

$M = \sqrt{(3^2 + 4^2)}$

Step 2: Calculate the squares.

$M = \sqrt{(9 + 16)}$

Step 3: Simplify.

$M = \sqrt{25}$

Final answer: M = 5.

If the edge threshold is 4, the pixel is classified as an edge because 5 > 4.

6. Thresholding

Thresholding separates pixels into foreground and background classes.

For dark defects:

$B(x,y) = 1$ if $I(x,y) < T$

$B(x,y) = 0$ if $I(x,y) \geq T$

7. Numerical Example

Assume the threshold value is $T = 100$.

Input intensity	Binary output	Interpretation
78	1	Possible defect
92	1	Possible defect
115	0	Background
151	0	Background

The binary output is:

[1, 1, 0, 0]

Here, 1 represents the suspected defect class, and 0 represents the background.

The display convention may be reversed depending on the image representation.

8. Improving Defect Detection

A single global threshold may fail when the metal surface has uneven illumination.

Adaptive thresholding can use local image statistics.

Formula:

$T(x,y) = \text{Local Mean}(x,y) - C$

Where C is a constant.

Additional morphological operations can improve the binary mask:

- Closing bridges small gaps in detected cracks.
- Opening removes small isolated foreground objects.
- Connected-component analysis groups connected pixels into candidate defects.

9. Complete Inspection Procedure

1. Capture the metal surface under controlled illumination.
2. Correct uneven illumination.
3. Apply light denoising.
4. Use Sobel or Canny edge detection.
5. Apply global or adaptive thresholding.
6. Refine the binary mask using morphological operations.
7. Measure crack length, width and area.
8. Validate detections against verified inspection results.

10. Image for Q4

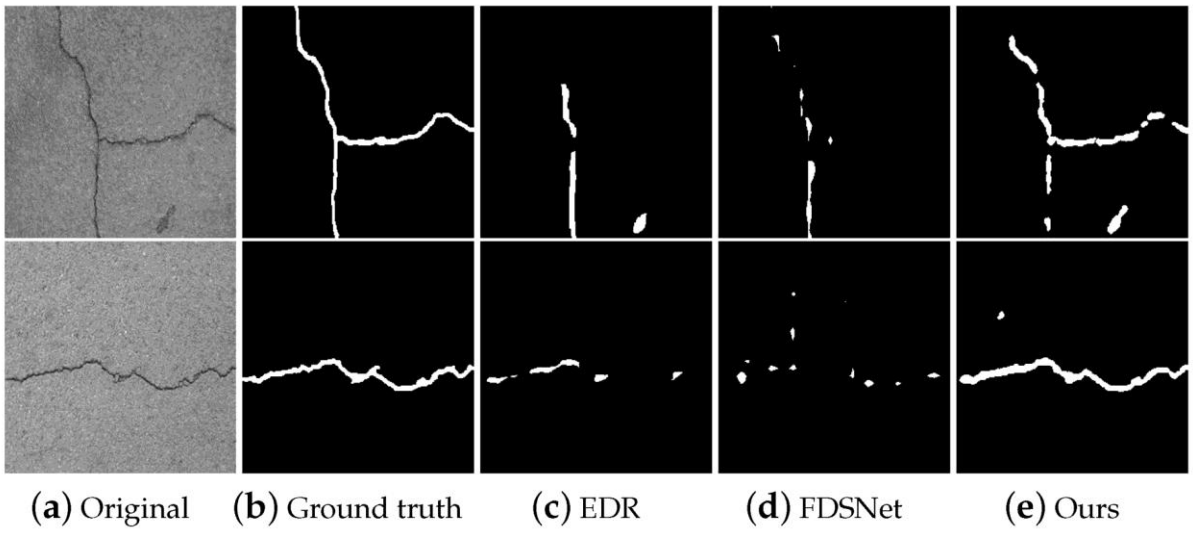


Figure 4: Illustration of original metal surface, detected edges and binary defect mask.

11. Conclusion

Edge detection identifies potential crack boundaries, while thresholding separates suspected defects from the background. Combining these techniques produces a binary defect image that supports automated quality inspection. Proper illumination correction and threshold selection are essential for reducing false detections.

Q5. Region-Based Segmentation and Boundary Detection

1. Problem Statement

An agricultural monitoring system uses aerial images to identify crop fields, water bodies and surrounding land regions. The images contain regions with similar intensity values and discontinuous boundaries. Region-based segmentation and boundary detection are applied to improve land-region identification.

2. Objective

1. To group neighbouring pixels with similar characteristics.
2. To identify agricultural fields and water bodies.
3. To connect discontinuous edges.
4. To improve land-region classification accuracy.

3. Theory

Region-based segmentation divides an image into connected regions based on similarity.

Pixels can be grouped according to:

- Intensity.
- Colour.
- Texture.
- Spectral information.
- Spatial connectivity.

Boundary detection identifies transitions between neighbouring regions.

Combining both techniques allows the system to identify coherent areas while preserving boundaries between different land types.

4. Region-Growing Segmentation

Region growing begins with one or more seed pixels.

Neighbouring pixels are added to a region when they satisfy a similarity condition.

Formula:

$|I(p) - \mu_R| \leq T$

Where:

- $I(p)$ = Intensity of the candidate pixel
- μ_R = Mean intensity of the region
- T = Similarity threshold

5. Step-by-Step Numerical Example

Suppose the mean intensity of a crop region is:

$\mu_R = 105$

Similarity threshold:

$T = 12$

Example 1: Candidate intensity = 110

Step 1: Calculate the difference.

$|110 - 105| = 5$

Step 2: Compare with the threshold.

$5 \leq 12$

Final result: The pixel can be included in the region if it satisfies connectivity and other segmentation conditions.

Example 2: Candidate intensity = 135

Step 1: Calculate the difference.

$|135 - 105| = 30$

Step 2: Compare with the threshold.

$30 > 12$

Final result: The pixel is not included in that region.

This process continues until no eligible neighbouring pixels remain.

6. Boundary Detection

Boundary detection identifies transitions between different regions.

The Sobel gradient magnitude is:

$M(x,y) = \sqrt{(Gx^2 + Gy^2)}$

Pixels with sufficiently high gradient values are marked as potential boundaries.

The Canny edge detector can also be used to produce thin, connected edges.

7. Edge Linking

In aerial images, boundaries may be discontinuous because of noise, shadows or similar intensity values.

Edge linking connects nearby edge pixels using:

1. Spatial distance.
2. Gradient direction.
3. Intensity similarity.
4. Boundary continuity.

Morphological closing can bridge small gaps. However, excessive closing may merge unrelated regions.

8. Combined Processing Workflow

Step 1: Image preprocessing

Normalize the aerial image and reduce noise.

Step 2: Initial segmentation

Use region growing, region merging or superpixel segmentation to identify similar areas.

Step 3: Boundary detection

Apply Sobel or Canny edge detection.

Step 4: Edge linking

Connect small discontinuities in the detected boundaries.

Step 5: Region refinement

Use the detected boundaries to guide region merging and correct segmentation errors.

Step 6: Classification

Assign labels such as crop field, water body and surrounding land.

Step 7: Evaluation

Compare the detected regions with manually annotated reference images.

9. Evaluation Using Intersection over Union

Segmentation quality can be measured using Intersection over Union (IoU).

Formula:

IoU = Area of Intersection / Area of Union

Suppose the intersection contains 80 pixels and the union contains 100 pixels.

Step 1: Substitute the values.

IoU = 80 / 100

Step 2: Calculate.

IoU = 0.8

Step 3: Convert into percentage.

IoU = 0.8 × 100 = 80%

Final answer: IoU = 80%.

A higher IoU generally indicates better agreement between the predicted region and the reference region.

10. Applications

Region-based segmentation and boundary detection are useful for:

1. Identifying crop fields.
2. Mapping water bodies.
3. Monitoring land-use changes.

4. Estimating agricultural area.
5. Detecting boundaries between neighbouring fields.
6. Supporting precision agriculture and environmental monitoring.

11. Image for Q5

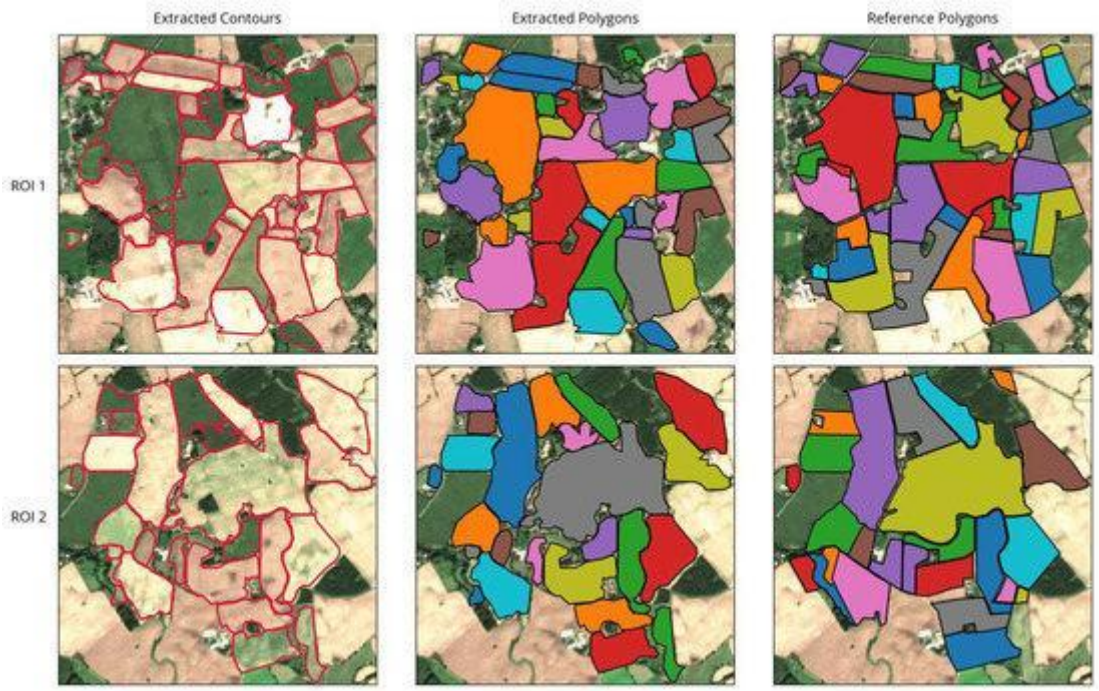


Figure 5: Illustration of original aerial image, segmented regions and detected boundaries.

12. Conclusion

Region-based segmentation groups neighbouring pixels with similar characteristics, while boundary detection identifies transitions between different land regions. Edge linking connects broken boundaries. Combining these techniques improves the identification of crop fields, water bodies and surrounding land regions in aerial images.